INCH-POUND
MIL-DTL-24640/25
9 October 2008

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, -20 °C TO + 125 °C, 600 VOLTS, TYPE 9XS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24640.

Construction, Non-Watertight

Components 1- 4:

First	-	Component wires shall be SAE-AS81044, except print identification shall be in accordance with table II. Testing shall be in accordance with the component wire ratings and requirements listed on this slant sheet. Conformance testing procedures shall be in accordance with SAE-AS81044.
Second	-	Two component wires twisted together. The lay length shall be in accordance with table I.

Component 5:

First	-	Component wires shall be SAE-AS81044, except identification shall be in accordance with table II. Testing shall be in accordance with the component wire ratings and requirements listed on this slant sheet. Conformance testing procedures shall be in accordance with SAE-AS81044.

Assembly:

Third	-	The components 1 - 5 shall be cabled together with a lay in accordance with table I. Fillers may be used as necessary to form a round cable.
Fourth	-	Binder tape applied helically with overlap.
Fifth	-	Braided shield of AWG 34 tin-coated copper. Braid angle of 40 to 50 degrees; minimum coverage of 90 percent.
Sixth	-	An optional binder tape.
Seventh	-	Cross-linked polyolefin jacket with a minimum wall thickness of 0.050 inch. Color black.

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TABLE I. Construction details.

Military Part No. M24640/25	Type & Size	Conductor Size AWG	Conductor Resistance	Component Wire	Pair Lay (inch)	Cable Lay (inch)	Minimum Bend Radius (inch)	Overall Diameter (inches)	
								Type 9XS	
								(min.)	(max.)
-01UO	9XS-12	12	2.02	SAE-AS81044/12	2.50±0.25	7.5±1.0	5	0.625	0.675
-02UO	9XS-10	10	1.26	SAE-AS81044/9	3.50±0.50	9.5±1.0	7	0.845	0.905
-03UO	9XS-8	8	0.701	SAE-AS81044/9	5.50±0.50	12.5±1.0	10	1.200	1.300

TABLE II. Color code.

Component	Insulation Color	Print
1	White	Black
	White	Brown
2	White	Black
	White	Red
3	White	Black
	White	Orange
4	White	Black
	White	Green
5	White	Black

COMPONENT WIRE REQUIREMENTS:

Qualification required.

Accelerated aging	Oven temperature 250±2 °C (482±3.6 °F) for 6 hours
Blocking	150±2 °C (302±3.6 °F)
Dielectric test	
Primary insulation	3 kilohertz spark test, 1500 volts (root mean square) 100%
Component wire	3 kilohertz spark test, 5500 volts (root mean square) 100%
Flammability	30 seconds (max.); 3 inches (max.); no flaming tissue paper
Humidity resistance	5000 megohms for 1000 feet, minimum insulation resistance after humidity exposure
Insulation resistance	5000 megohms for 1000 feet (min.)
Life cycle	Oven temperature 200±2 °C for 168 hours
Physical properties of primary insulation	
Tensile strength	2500 pounds per inch (min.)
Elongation	150 percent (min.)

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Shrinkage	0.125 inch (max.) at 250±2 °C (482±3.6 °F)
Surface resistance	500 megohms inches (min.), initial and final readings
Thermal shock	Oven temperature, 150±2 °C (302±3.6 °F) Change in measurement: 0.060 inch (max.)
Wet dielectric test	2500 volts (root mean square)
Wrap test	Mandrel wrap test required

EXAMINATION AND TESTS:

Basic Electricals:	
Conductor resistance (ohms/1000 feet at 25 °C, max.)	See table I
Voltage withstand (volts, root mean square, min.)	
Conductor to conductor	2000
Conductor to shield	1500
Insulation resistance (megohms/1000 feet, min.)	
Conductor to conductor	2500
Conductor to shield	2500
Conductor and shield continuity	No Failure
Jacket flaws	No Failure
Group A:	
Visual and dimensional	No Failure
Group B:	
Cross-linked proof test (percent, max.)	
Jacket	50
Cold bending, cable mandrel diameter 12 times cable diameter	No Damage
Tear (pounds per inch thickness, min.)	35
Physicals (unaged)	
Jacket (cable)	
Tensile strength (lb/in^2, min.)	1300
Elongation (percent, min.)	160
Group C:	
Physicals (aged)	
Air oven	
Jacket (cable)	
Tensile strength (percent of unaged, min.)	60

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Elongation (percent of unaged, min.)	60
Heat distortion (percent of unaged, max.)	30
Permanence of printing (insulation) (cycles, min.)	75
Permanence of printing (jacket) (cycles, min.)	125
Shrinkage	No Failure
Shield (conformance to material, construction, and coverage)	No Failure
Group D:	
Flame propagation (cable)	No Failure
Qualification Inspection:	
Qualification inspection shall include basic electricals, groups A, B, C, and D, plus the following:	
Aging and compatibility (cable) (125±5 °C)	No Failure
Abrasion resistance (jacket) (scrapes, min.)	75
Acid gas equivalent (percent, max.)	
Jacket	2
Fillers	2
Insulation	18
Halogen content (percent, max.)	
Jacket	0.2
Fillers	0.2
Smoke index, max.	
Jacket	25
Fillers	45
Insulation	35
Toxicity index, max.	
Jacket	5
Fillers	5
Insulation	1.5
Immersion (jacket)	
Tensile strength (percent of unaged, min.)	50
Elongation (percent of unaged, min.)	50
Durometer (jacket) [(Type A) hardness, min.]	80
Weathering (jacket)	No Failure

UNIT ORDERING LENGTHS: All sizes 250 feet (nominal)

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Custodians:
Army – MI
Navy – SH

Preparing Activity:
Navy – SH
(Project 6145-2008-115)

Review Activities:
Army –AR, AV, CR, CR4
Navy – CG, EC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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